

Kian Sahafi

Software Engineer

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Work Experience

- Aug 2025 – May 2026 **Software Engineer, YUUNIQ Health, Winterthur, Zurich, Switzerland**
- Led the migration from a non-SQL database to an SQL based database
 - Transformed a traditional LLM-based chatbot into an agentic system with persistent memory features.
 - Created a Recommender system for content recommendation based on user interaction.
 - Implemented the company's payment module, integrating secure and reliable transaction workflows.
 - Diagnosed and resolved a production failure in the application: restoring the relevant service within [$<1h$] and stabilizing uptime to 98%.
 - Optimized ChatBot context handling and aggregated recent interactions: reducing tokens by 60%.
 - Increased test coverage across the project by 30%
 - Designed and executed the merger integration plan with another company
- Apr 2022 – Sep 2024 **Software Engineer - Co-Founder, Firozak Co., Zanzan, Iran**
- Developed 9 automation bots for data processing and retrieval across social media platforms.
 - Partnered with stakeholders to identify growth opportunities and resolve customer pain points.
 - Pitched the company's solutions to prospective clients, contributing to a 50% user increase in the first year.
 - Reviewed code, debugged issues, and mentored new employees.
- Apr 2021 – Apr 2022 **Software Engineer, Tose'e System Tavana, Zanzan, Iran**
- Implemented a dynamic barcode generator supporting multiple formats (C#, Angular).
 - Refactored and redesigned the front-end architecture, halving bundle size and improving load times.
 - Wrote unit and end-to-end tests achieving 75% coverage, reducing incidents by 24%.
 - Mentored new joiners on the ERP warehouse team and collaborated with domain experts to shape feasible solutions.

Education

- 2024–Now **MSc, Computer Science, University of Bern, Switzerland**
- **Master thesis:** *Design and Evaluation of a Reference Architecture for Modular, Risk-Aware Backtesting Systems.* Designed and evaluated a modular architecture for historical trading simulation, comparing batch and event-driven pipelines.
 - Collaborated on "Universal Merge: A Language-Agnostic Recommender for Merge Conflict Resolution" with Prof. Timo Kehrer.
- 2020–2024 **BSc, Computer Engineering, University of Zanzan, Iran**
- Top team in university ICPC contest.
 - Teaching Assistant for Advanced Programming and other courses.
 - Member, University Innovation Center Incubator.
- 2016–2020 **High School, Shahid Beheshti Exceptional Talents High school, Iran, Mathematics and Physics**
- Among Top 10 Students in Mathematics and Physics Department.
 - educated in the school of NODET(National Organization for Development of Exceptional Talents)

Research Experience

- Oct 2024 – Aug 2025 **Research Assistant, Software Engineering Group (SEG), University of Bern, Bern, Switzerland**
- Supervisor: Prof. Timo Kehrer.
 - Project: Merge++ : work on automated merge conflict resolution in GitHub repositories.
 - Conducted literature review, implemented research prototypes, and contributed to experimental design and evaluation.
 - Built/maintained reproducible experiment pipelines and documented results for internal reports.
seg.inf.unibe.ch

Publications

- Pending... Universal Merge: A Language Agnostic Recommender for Merge Conflict Resolution
- Feb 2024 "A Hybrid Deep Learning-based Model for Off-target Prediction in CRISPR Systems," submitted

Projects

- Jun 2026 – **Caritas Dashboard**, *Bern University of Applied Sciences (BFH)*
Present Built a fully local offline analytics platform using Python, PostgreSQL, Apache Superset, and Podman, with CSV/Excel ingestion and automatic type inference.
- Dec 2024 **Oxford-IIIT Pet Dataset Classification (AI)**
Transfer learning with ResNet-50.
○ Code: github.com/kiansahafi/PetBreedClassifier-ResNet50
- Oct 2023 **Object Detection Project - AI**
Traffic Sign Detection with YOLO
○ You can see the Project on My GitHub: [AI-Traffic-sign-Detection](#)
- Dec 2022 **Compiler Project**
Compiler Project Using Python
○ You can see the Project on My GitHub: [Compiler Project University](#)

Awards & Honors

- Sep 2024 Acknowledged in DTMP-Prime: A deep transformer-based model for predicting prime editing efficiency and PegRNA activity for creating the website and managing the GitHub page.
- Feb 2024 Top-ranking final project submission in Computer Engineering Faculty - Autumn Semester 2024.
- 2023 Ranked Among the Top 25 ICPC Contest.
- 2019 Ranked 2206 Among 164,000 participants in the national university entrance exam for B.Sc. degree.

Skills

Languages	Python, Java, C#, JavaScript	Databases	MS SQL Server, MySQL
Frameworks	AI (TensorFlow, Keras), Web (Angular, React, .NET, Django)	Operating Systems	Linux, macOS

Additional Attended Courses

- Feb 2022 Domain Driven Design Software Architecture Course, Tehran (dddiran.com)

Teaching Assistant

- 2023 Spring **Advanced Programming**, (*Dr.Hossein Mohammadi*), University of Zanjan, Iran
○ Mentored and assisted students in small groups.
○ organized troubleshooting classes, and designed and delivered the final project for the class.
- 2022 Fall **Advanced Programming**, (*Dr.Saeed Rahmani*), University of Zanjan, Iran
○ Head of the TA group and managed the TA Group.
○ organized the managed course project.
- 2022 Fall **System design & analysis**, (*Dr.Saeed Rahmani*), University of Zanjan, Iran
○ Provided guidance and mentoring to students in small groups.
○ Assisted in the Project Delivery Process.
- 2021 Spring **Advanced English for Computer Students**, (*Dr.Leila safari*), University of Zanjan, Iran
○ Provided assistance to students in exercises delivery and individualized help.

Languages

Persian	Mother tongue
Turkish	C1
English	C2 (IELTS 7.0)
German	B1
French	A1